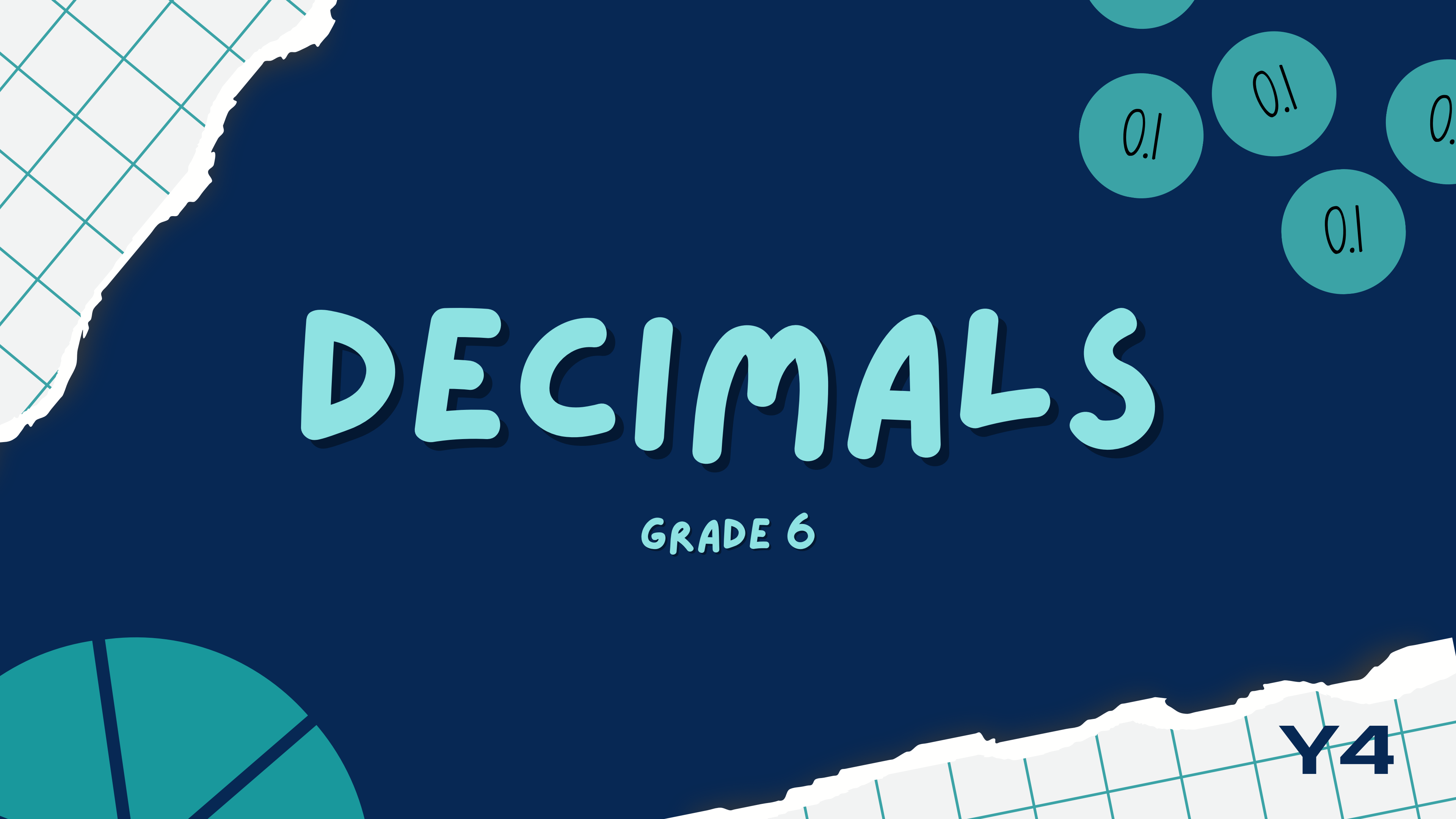
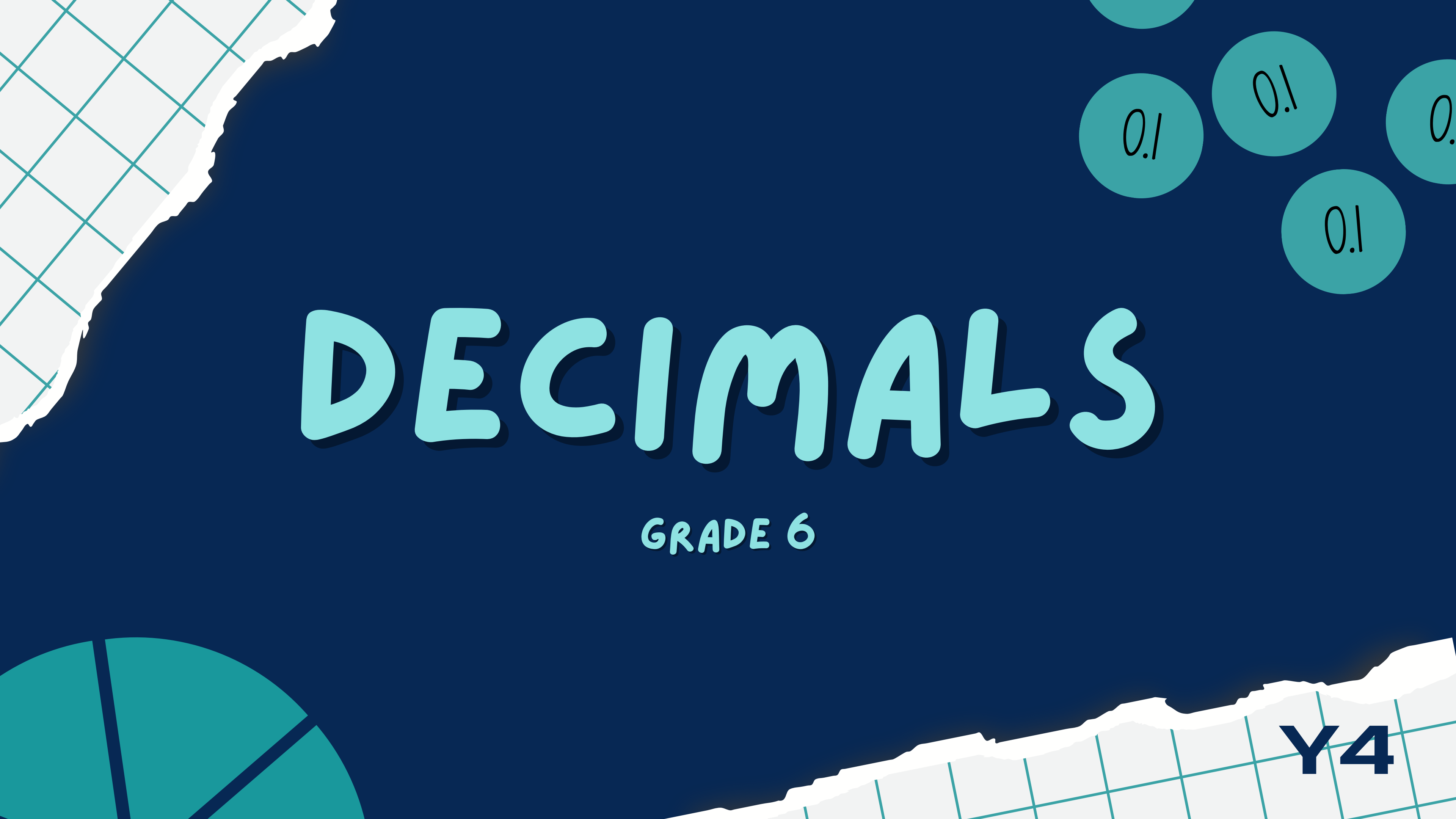




DECIMALS

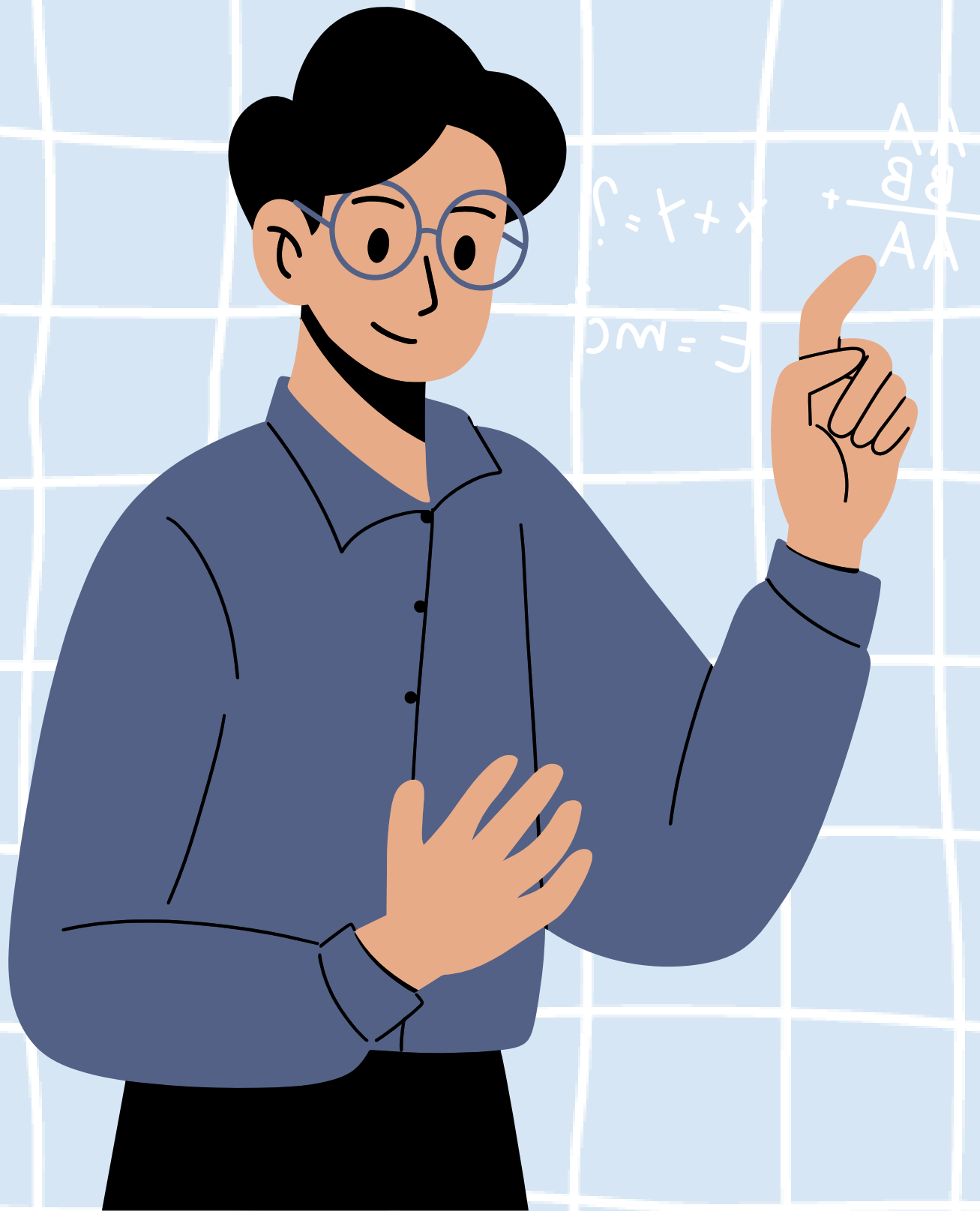
GRADE 6



Y4

Introduction

Where have you seen
decimals in real life?



REAL-LIFE APPLICATIONS

Fractions and decimals are used in:

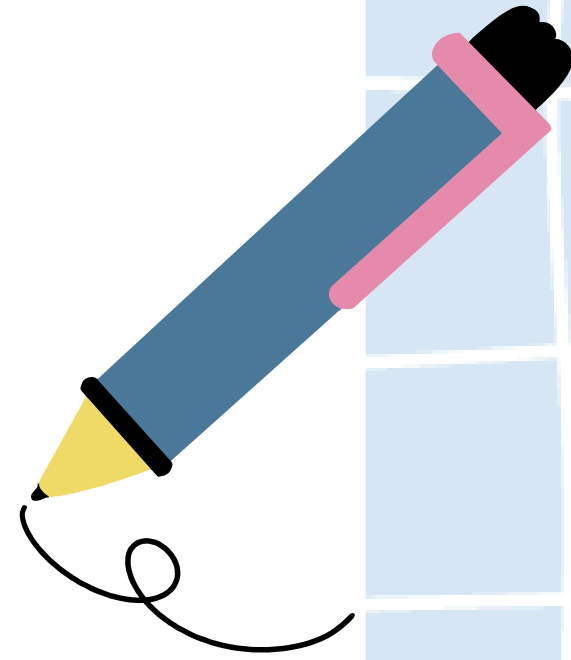
1. Money and Shopping

- Prices often use decimals: e.g., \$2.50, \$9.99
- Discounts use fractions: “Half price” = $\frac{1}{2}$
- Example: A \$20 shirt with 25% discount $\rightarrow$ fraction $\frac{1}{4} \rightarrow$ pay \$15.

2. Cooking and Recipes

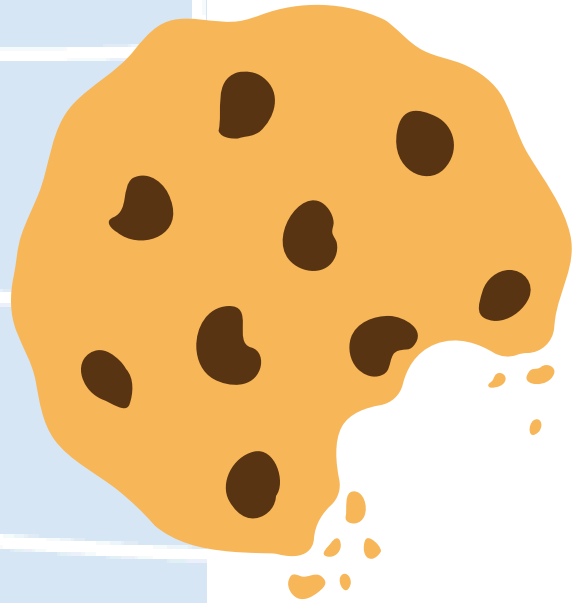
- Ingredients are in fractions: $\frac{1}{2}$ cup of sugar, $\frac{3}{4}$ teaspoon of salt.
- You may need to convert them to decimals when adjusting recipes.
- Example: 0.25 liters of milk = $\frac{1}{4}$ liter

Decimals



Decimals are numbers that have a **whole part** and a **part of a whole**, **separated by a decimal point**. They are used for money transactions, weight, measurements, test score or sport scores.

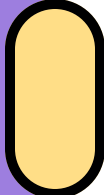
Example



If you have 10 cookies and you ate 0.3 of them, that means you ate 3 cookies because 0.3 means 3 tenths, and 3 tenths of 10 is 3.

$$0.3 = 3/10$$

$$3/10 \text{ of } 10 \text{ cookies} = 3 \text{ cookies}$$

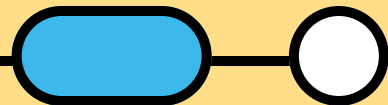
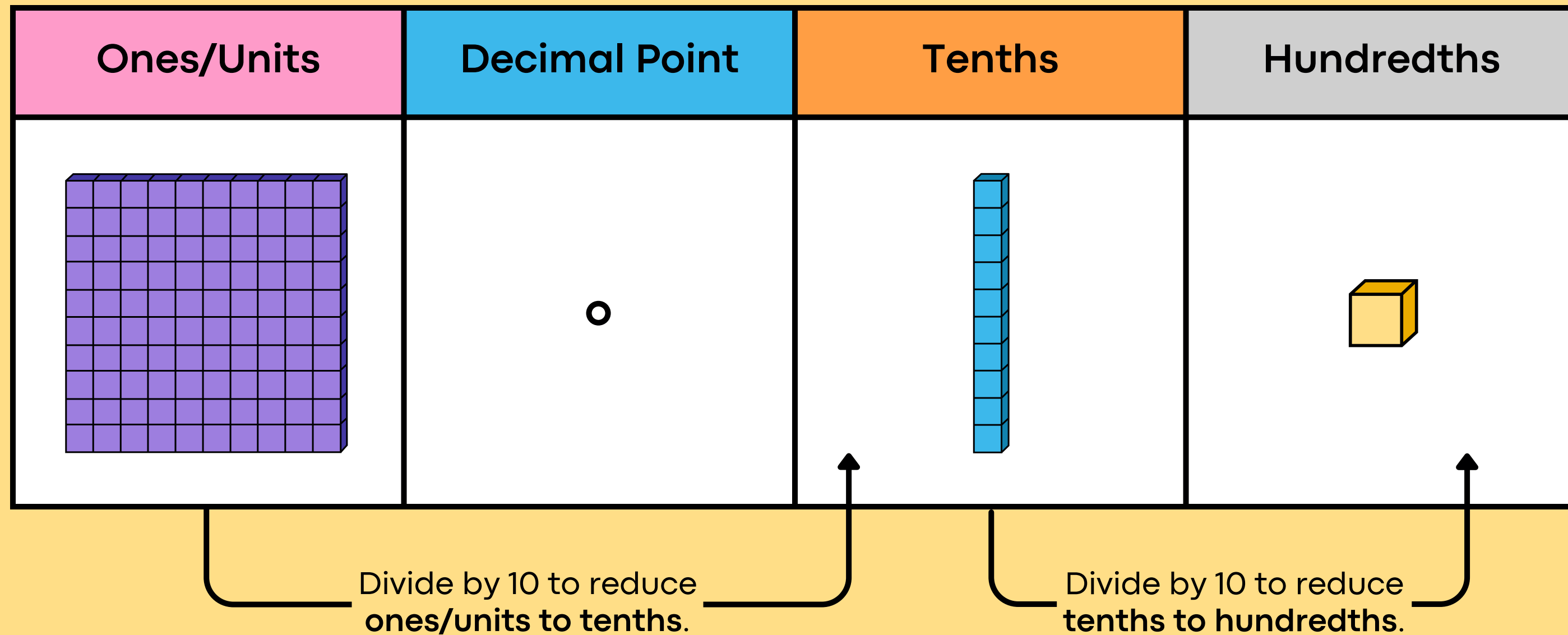




The place value of a digit is
important when comparing
decimal numbers.

Remember the values in decimals:

As the place value moves to the right, the value decreases by a tenth.



Which decimal is greater?

0.30

or

0.3



Ones/Units	Decimal Point	Tenths	Hundredths
0	.	3	0

Ones/Units	Decimal Point	Tenths	Hundredths
0	.	3	

They have the same value.

If you only have
0.75 coins, which
snack can you buy?

Donut

Muffin

Your name won't be shared. Never submit
passwords. | 0 votes

0.90



0.70

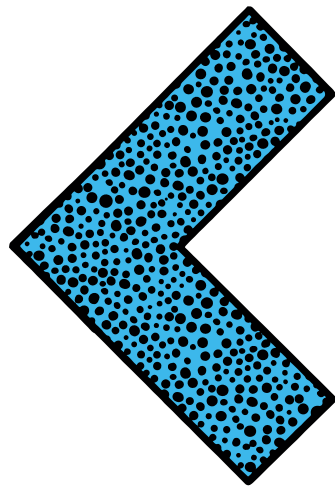


You can only buy a **muffin** since 0.70 is less than 0.75.

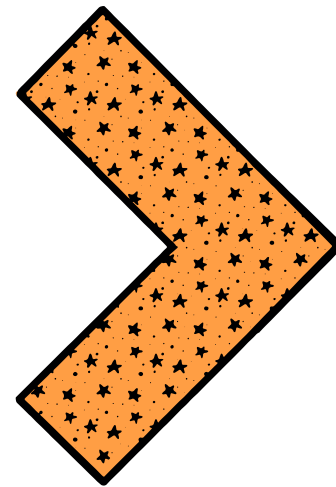
Let's learn more on the next slides!



**We can use symbols
to show the relationship
between two numbers.**



means
less than



means
greater
than

Is 0.80 less than or greater than 0.76?

Less than

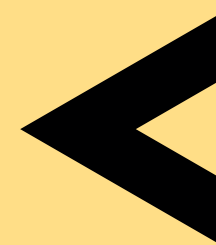
Greater than

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0.80

?

0.76



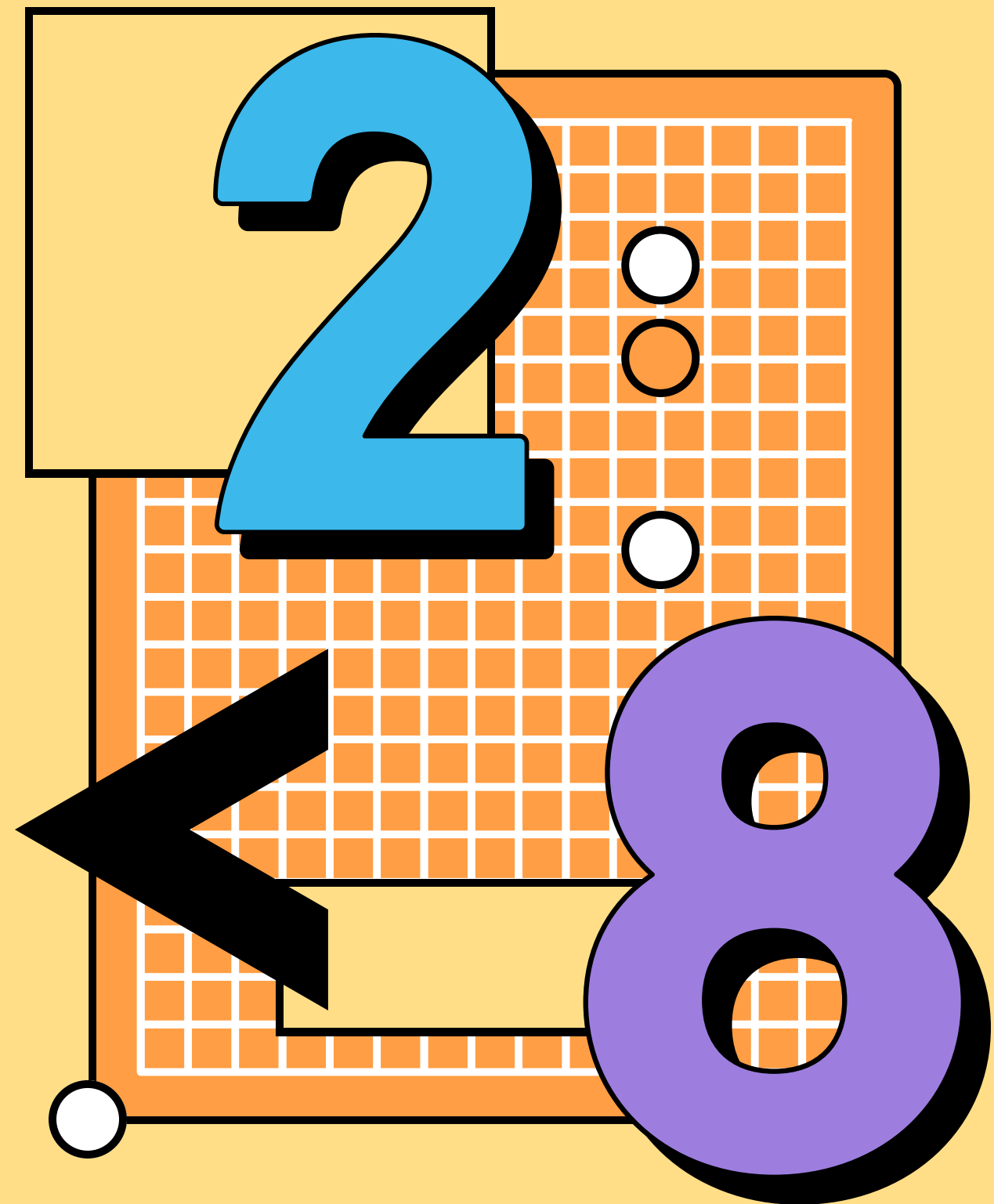
or



Less than or greater than

$$0.80 > 0.76$$

By comparing 8 and 7, we know that 8 is **greater than** 7.



Remember

Look at the place values, the further the digit is to the left, the greater its value.

TRY THIS!

Which food would
need the least amount
of time to prepare?

Tacos

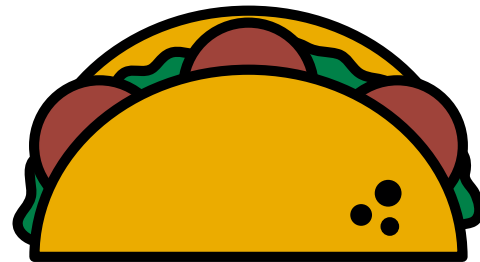
Pancakes

Pizza

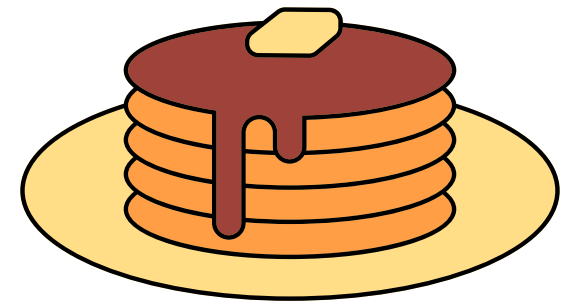
Submit

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1.25 hrs



1.20 hrs

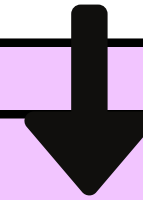


1.55 hrs



DECIMAL

decimal is another way of writing fractions, especially those with denominators like 10, 100, 1000, etc.



Example:

$$0.5 = \frac{5}{10} = \frac{1}{2}$$

Place values:

Tenths (0.1), Hundredths (0.01), Thousandths (0.001)

FRACTIONS TO DECIMALS

a fraction to a decimal means dividing the numerator (top number) by the denominator (bottom number).



Examples:

$$\begin{array}{l} \textcircled{1} \quad \frac{1}{2} = 1 \div 2 = 0.5 \\ \textcircled{2} \quad \frac{3}{4} = 3 \div 4 = 0.75 \end{array}$$

DECIMALS TO FRACTIONS

a decimal to a fraction means writing the decimal number as a fraction with denominator 10, 100, 1000.

Example:

$$\textcircled{1} \quad 0.5 = \frac{5}{10}$$

SIMPLIFY $\rightarrow \frac{1}{2}$

Example:

$$\textcircled{2} \quad 0.75 = \frac{75}{100}$$

SIMPLIFY $\rightarrow \frac{3}{4}$

COMPARING FRACTIONS AND DECIMALS

To compare fractions and decimals, we need to change both into the same form (either both as decimals, or both as fractions).

If both are decimals → compare directly.
If both are fractions → compare by finding common denominators.

Example 1: Compare $\frac{1}{2}$ and 0.6

- ① Change $\frac{1}{2}$ into decimal ② Compare: $0.5 < 0.6$

$$\frac{1}{2} = 0.5$$

$$\checkmark \frac{1}{2} < 0.6$$

Rounding decimals

Look at the digit right after the place you're rounding to:

If that digit is 5 or more, round up

If it's less than 5, round down

Rounding to the nearest tenth:

$0.18 \rightarrow 0.2$ (8 is 5 or more)

$0.12 \rightarrow 0.1$ (2 is less than 5)

$0.15 \rightarrow 0.2$ (5 is 5 or more)

Add & subtract

1. Line up the decimal points.
2. If a number has fewer decimal places, add zeros to match.
3. Then add or subtract like normal.

$$4.236 + 3.450 = 7.686$$

$$6.00 - 0.72 = 5.28$$

$$0.17 + 0.08 = 0.25$$

$$1.89 - 1.70 = 0.19$$



QUESTION 1

$$27.43 + 18.68 = ?$$

Estimation:

Working out:

QUESTION 2

$$64.75 - 28.96 = ?$$

Estimation:

Working out:

QUESTION 3

$$31.98 + 22.41 = ?$$

Estimation:

Working out:

QUESTION 4

$$17.87 - 9.66 = ?$$

Estimation:

Working out:

QUESTION 5

$\$101.56 + \$235.91 = ?$

Estimation:

Working out:

QUESTION 6

$\$231.01 - \$105.99 = ?$

Estimation:

Working out:

QUESTION 7

$$\$623.20 + \$221.00 = ?$$

Estimation:

Working out:

QUESTION 8

$$\$411.09 - \$108.61 = ?$$

Estimation:

Working out:

QUESTION 9

$$8241.11 + 312.19 = ?$$

Estimation:

Working out:

QUESTION 10

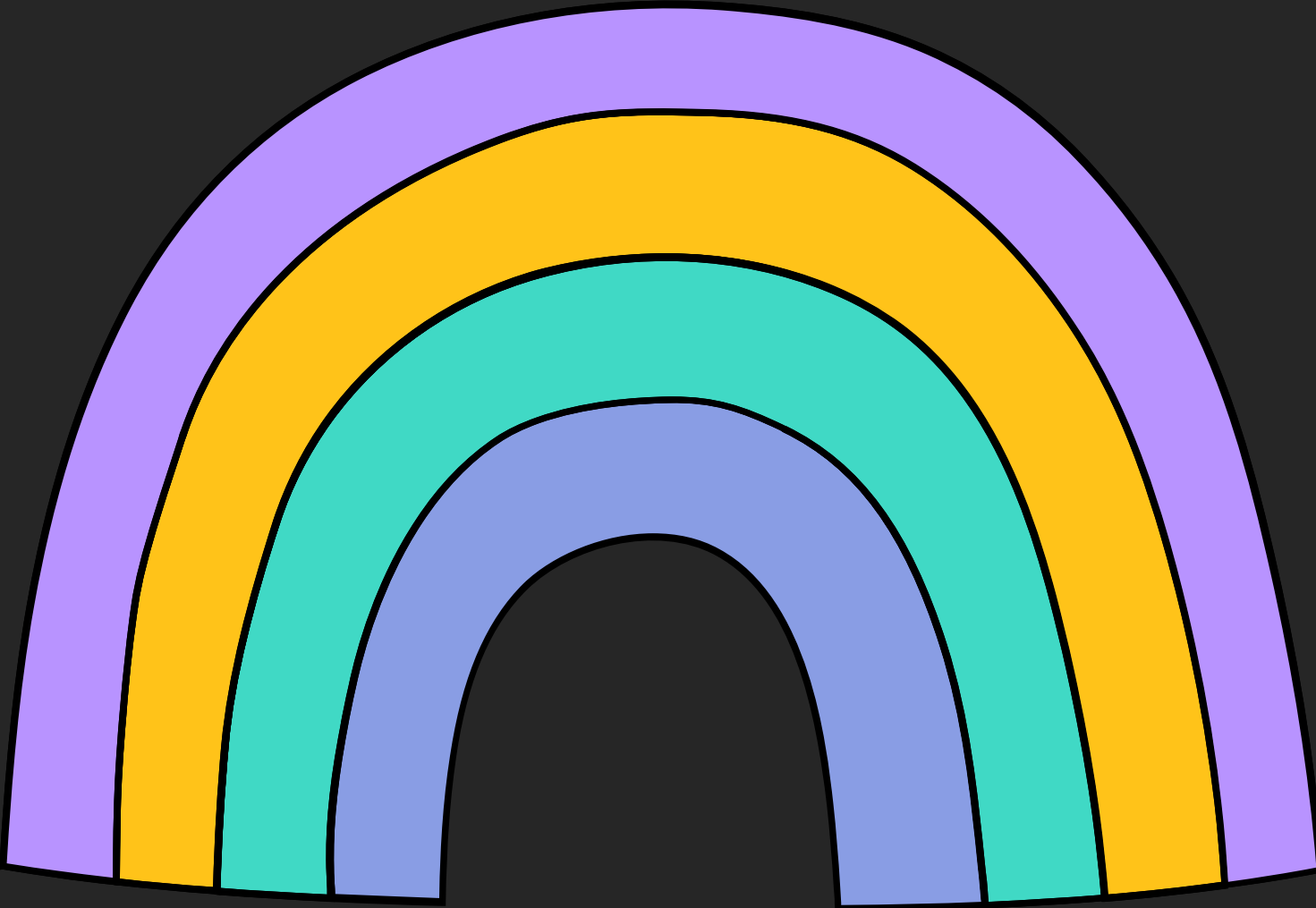
$$2243.01 - 122.20 = ?$$

Estimation:

Working out:

SOLUTIONS

1	46.11
2	35.79
3	54.39
4	8.21
5	\$337.47
6	\$125.05
7	\$844.20
8	\$302.48
9	8553.30
10	2120.81



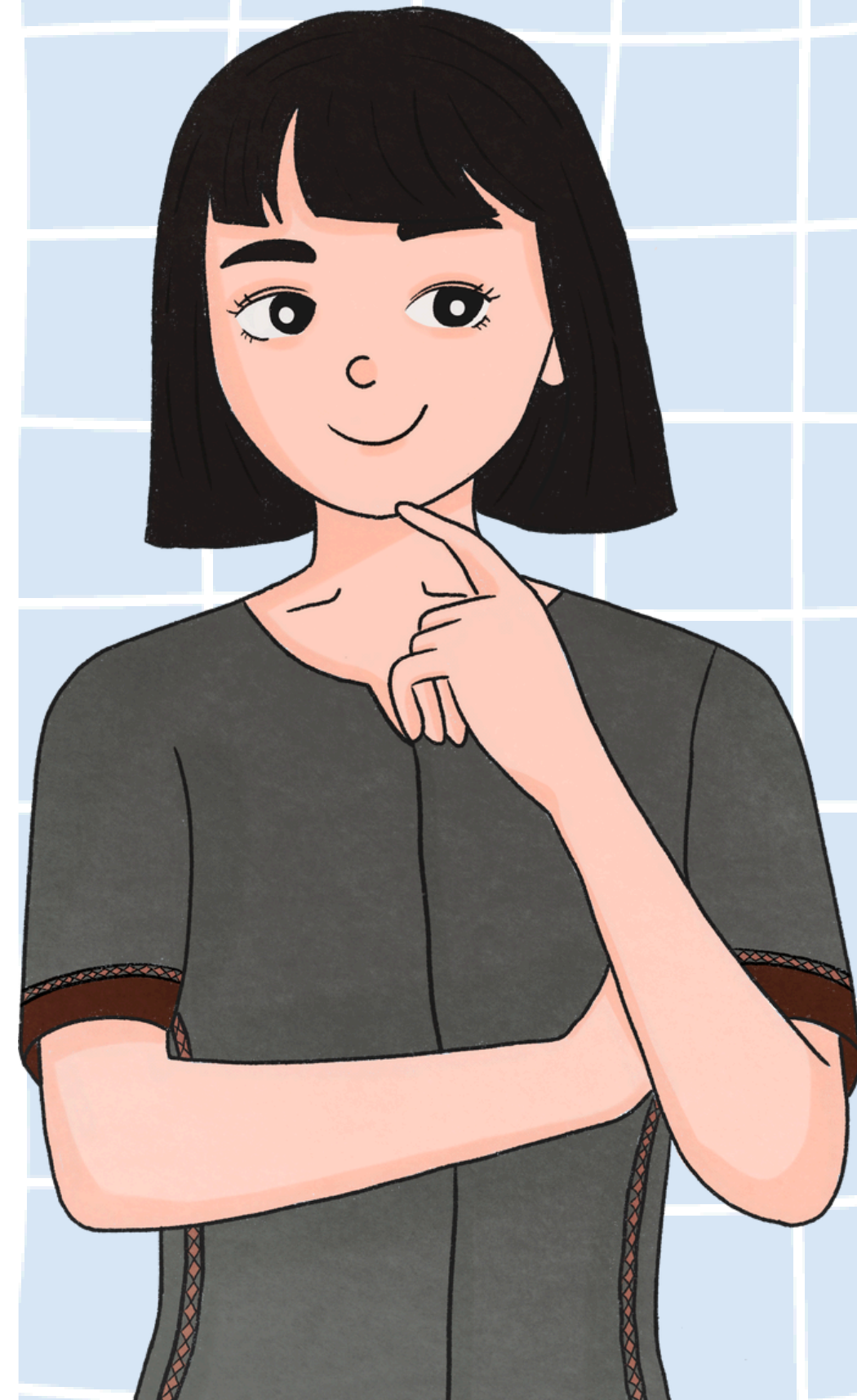
Multiply & divide

1. Count decimal places.
2. Multiply like whole numbers.
3. Move the decimal in the answer.

$$1.47 \times 0.2 \rightarrow 294 \rightarrow 0.294$$

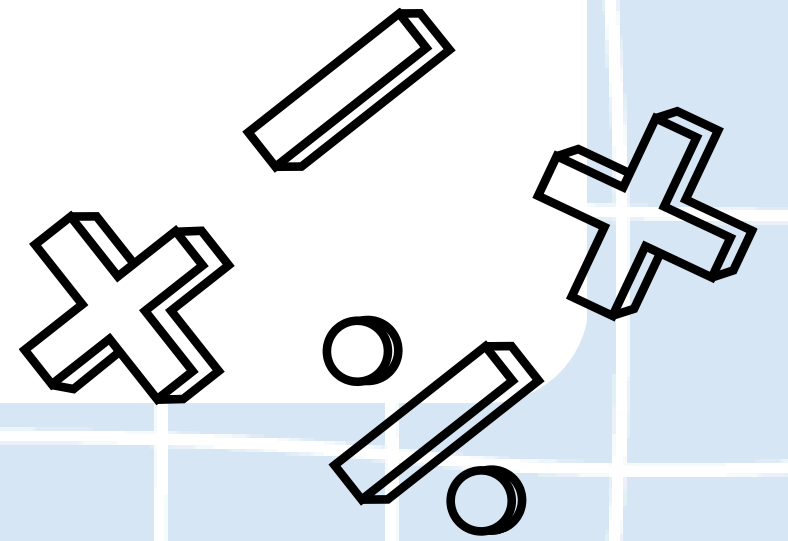
- ÷
1. Write as a fraction.
 2. Multiply top & bottom to remove decimal
 3. Divide.

$$0.2 \div 0.8 = 2/8 = 0.25$$



$$E = C^2$$

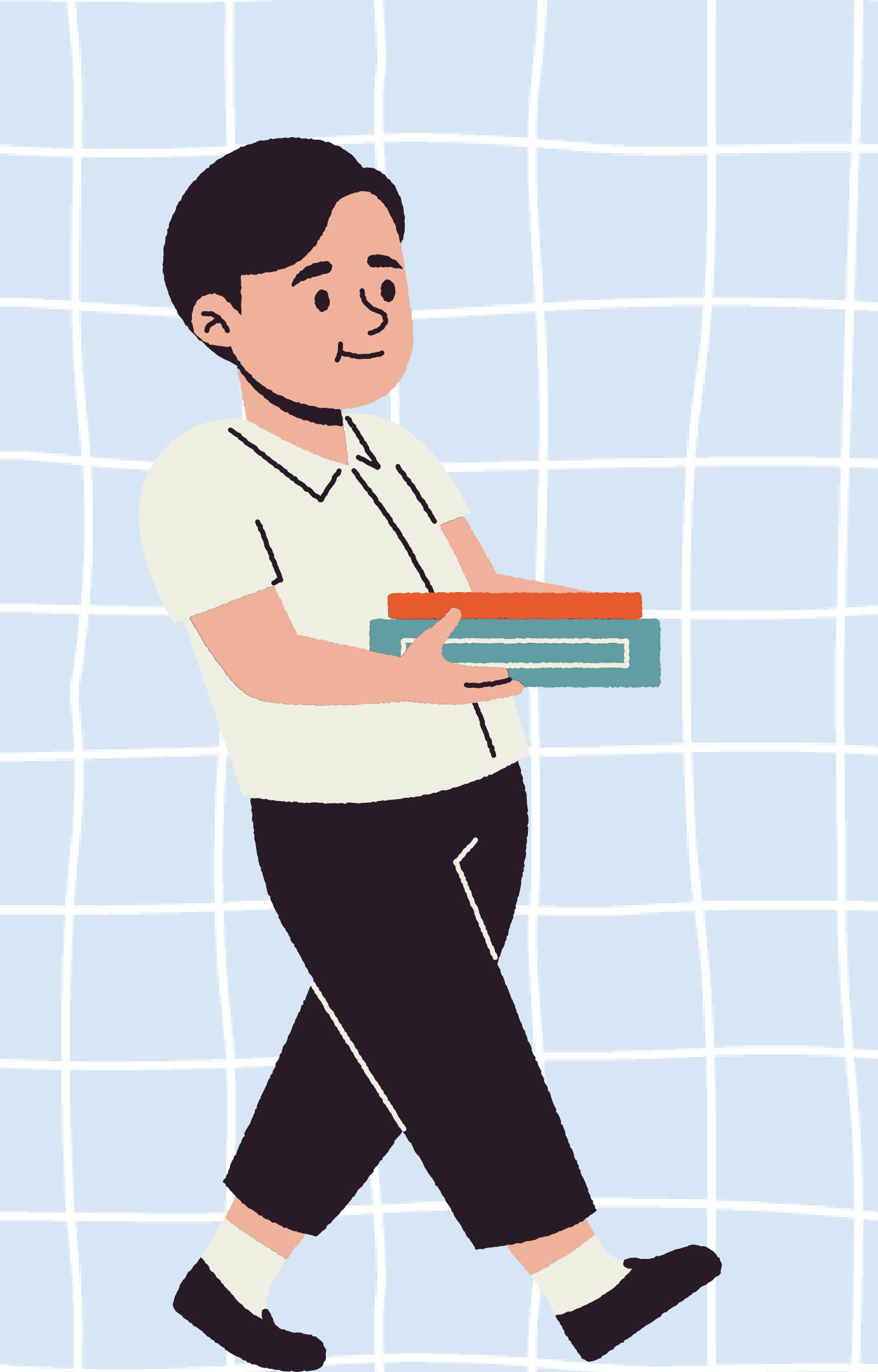
Quiz Time



Question #1

- A) Convert 2.4 to a percent
- B) $5.4/3$
- C) A water bottle is filled to 0.65 of its total capacity.
What percent is it?





Question #2

How are percents,
fractions, and
decimals related?



**Thank
You**